

Lecture 12

The Normal Distribution, z-scores and Transformations

What you should know: kinds of study

Experimental study

Units are assigned to treatments at random: the study creates the groups.

Observational study

Variables are measured on each unit and nothing is assigned.

Survey

Selected units answer questions, to describe a population.

What you should know: sampling designs

Simple random sample (SRS), and systematic sample

SRS: every sample of n is equally likely. Systematic: random start, then every k th unit.

Stratified random sample, and cluster sample

Stratified: an SRS from every stratum. Cluster: randomly chosen whole clusters.

Convenience sample, and voluntary response sample

The easiest to reach, or self-selected. Both are biased at any sample size.

What you should know: experimental designs

Completely randomized design

Every unit is assigned to a treatment completely at random.

Randomized block design

Random assignment is done separately within blocks of similar units.

Matched pairs design

Blocks of two similar units, or one subject who receives both treatments.

What you should know: terminology

The language of an experiment

Units, factor, levels, treatments, control group, placebo, blinding.

Association is not causation

A lurking variable can create an association with no causal link.

The two things randomness buys you

Random sampling lets you generalize. Random assignment lets you claim cause.

Data: female student heights

A survey of 379 college students recorded each student's height and gender; 262 are women.

X = the height of a female student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
...	...

n = 262 women (the file is sorted from shortest to tallest)

Data: Central Park in January

The National Weather Service has recorded the mean temperature in Central Park, New York, every month since 1869.

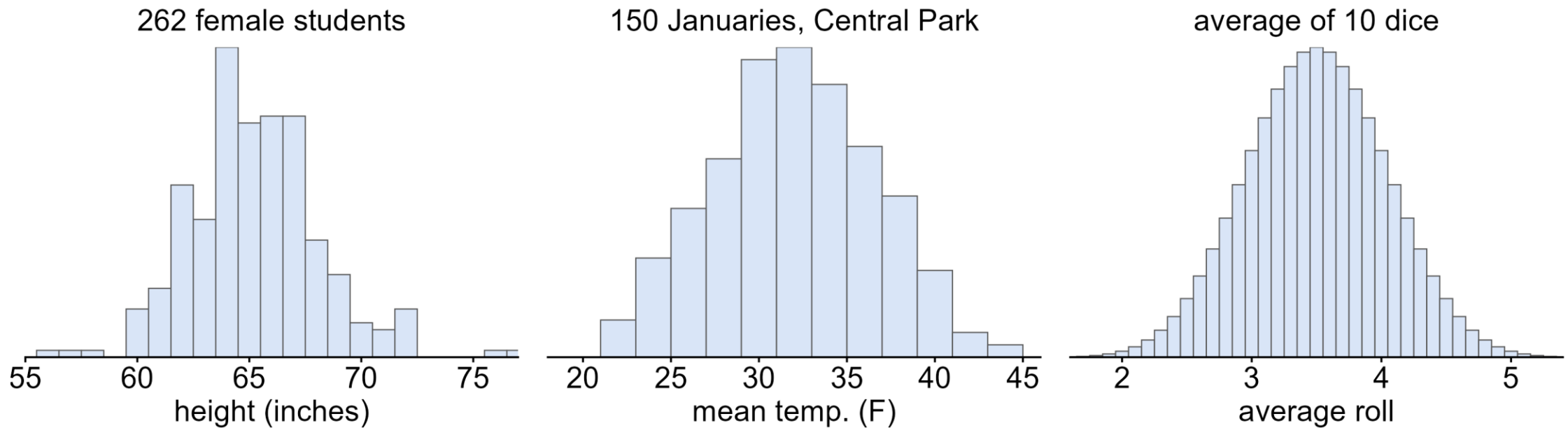
X = the mean January temperature in Central Park in a given year, in °F

YEAR	JAN	FEB	MAR	...	DEC	ANNUAL
1869	35.1	34.5	34.8	...	34.7	51.4
1870	37.5	31.3	34.1	...	34.1	53.6
1871	28.3	30.2	44.2	...	29.2	51.1
1872	28.8	29.9	30.5	...	26.7	51
1873	28.6	29.5	35.7	...	36.5	51
...	...	...	...	...	...	...

n = 150 years, 1869 to 2018

What do these have in common?

What do these three distributions have in common?

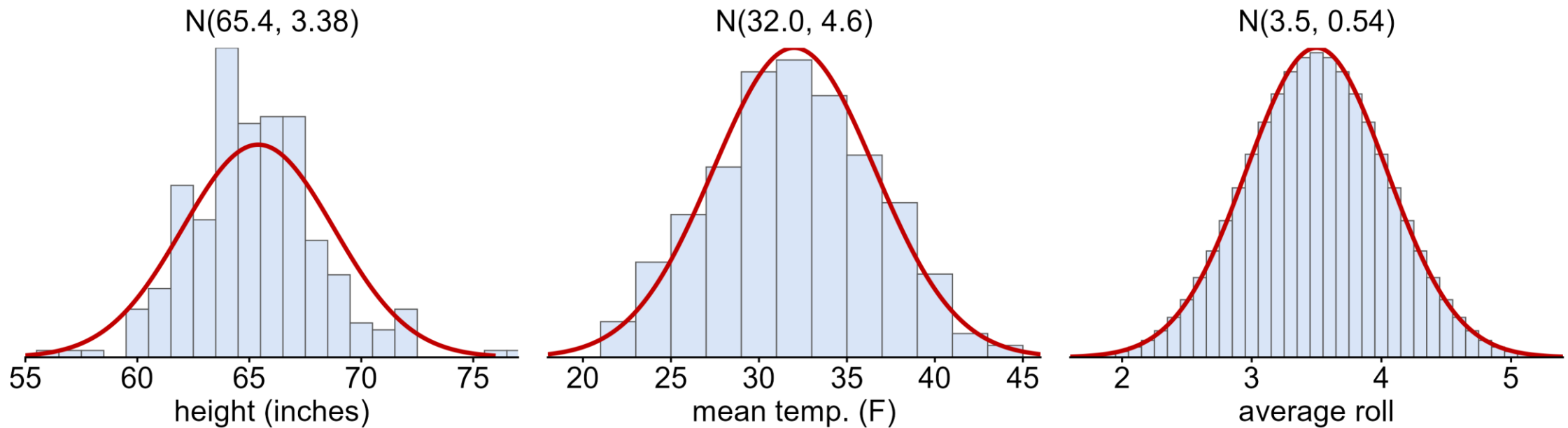


Each is single-peaked, symmetric and bell shaped

The normal distribution

normal distribution

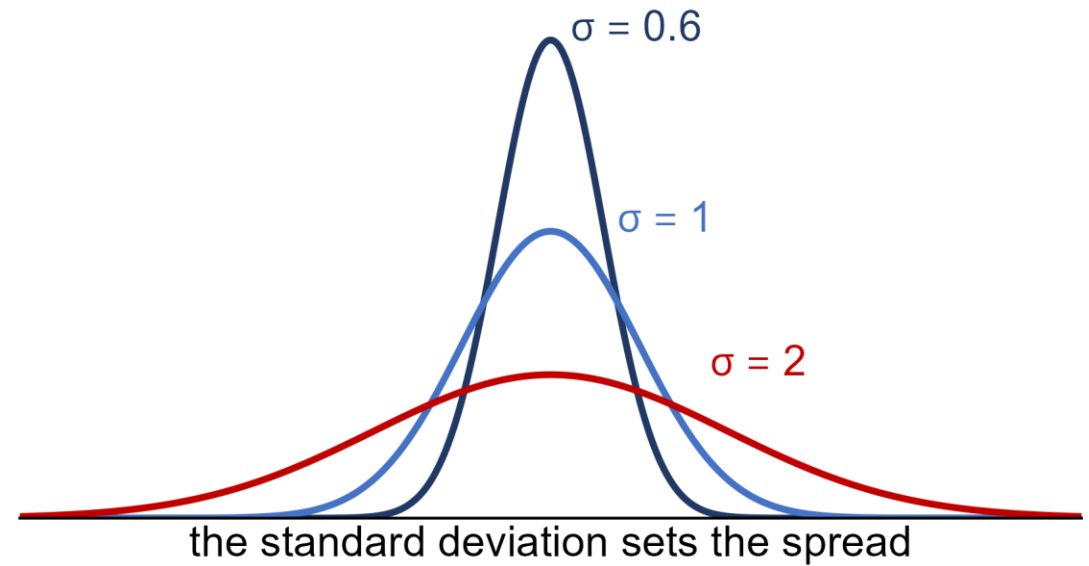
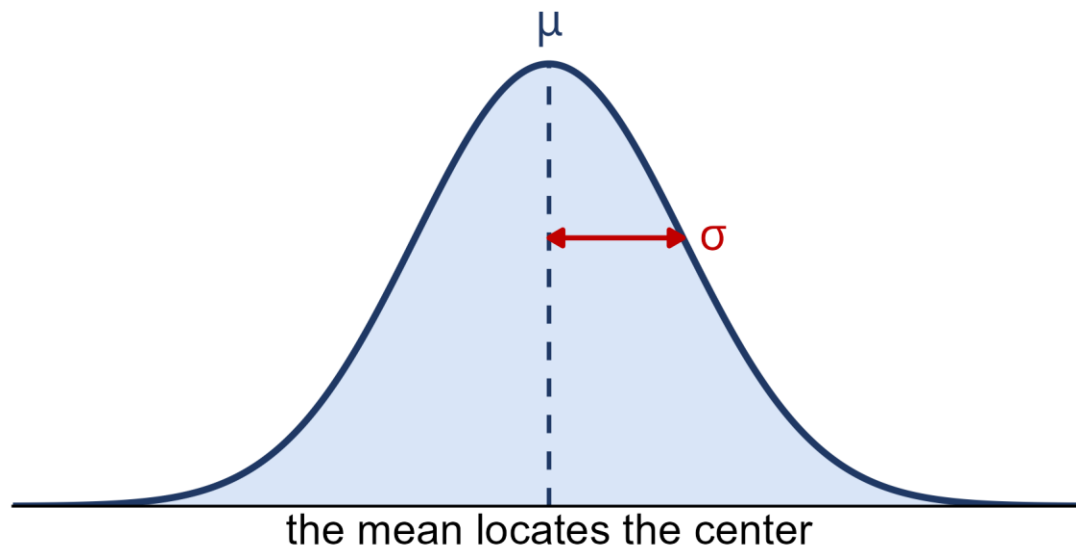
a symmetric, bell-shaped curve with its peak at the mean



Two parameters: μ and σ

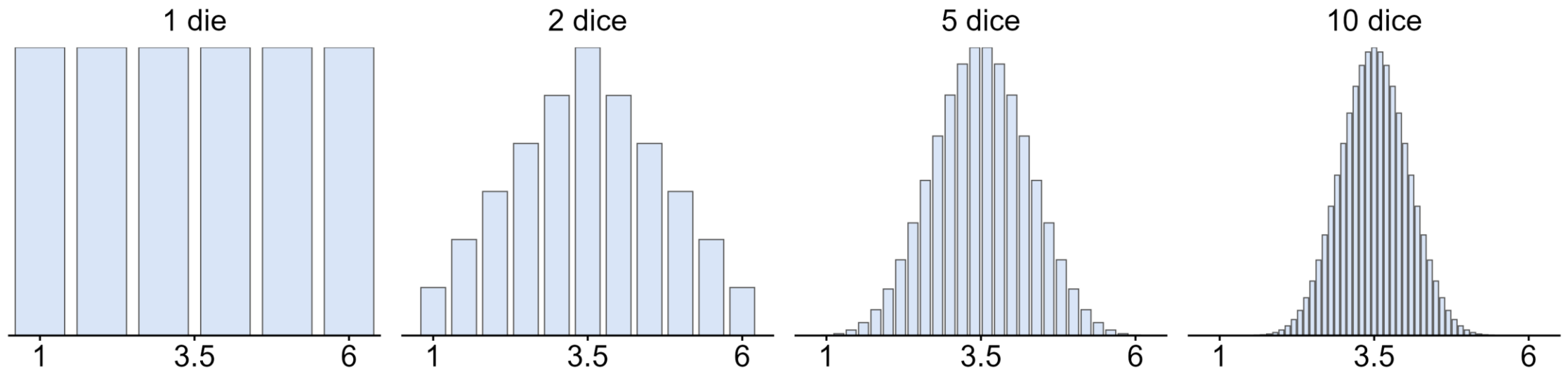
$$X \sim N(\mu, \sigma)$$

μ locates the center; σ sets the spread



Why is the normal so important?

Why does one curve matter so much?



Many measurements are approximately normal

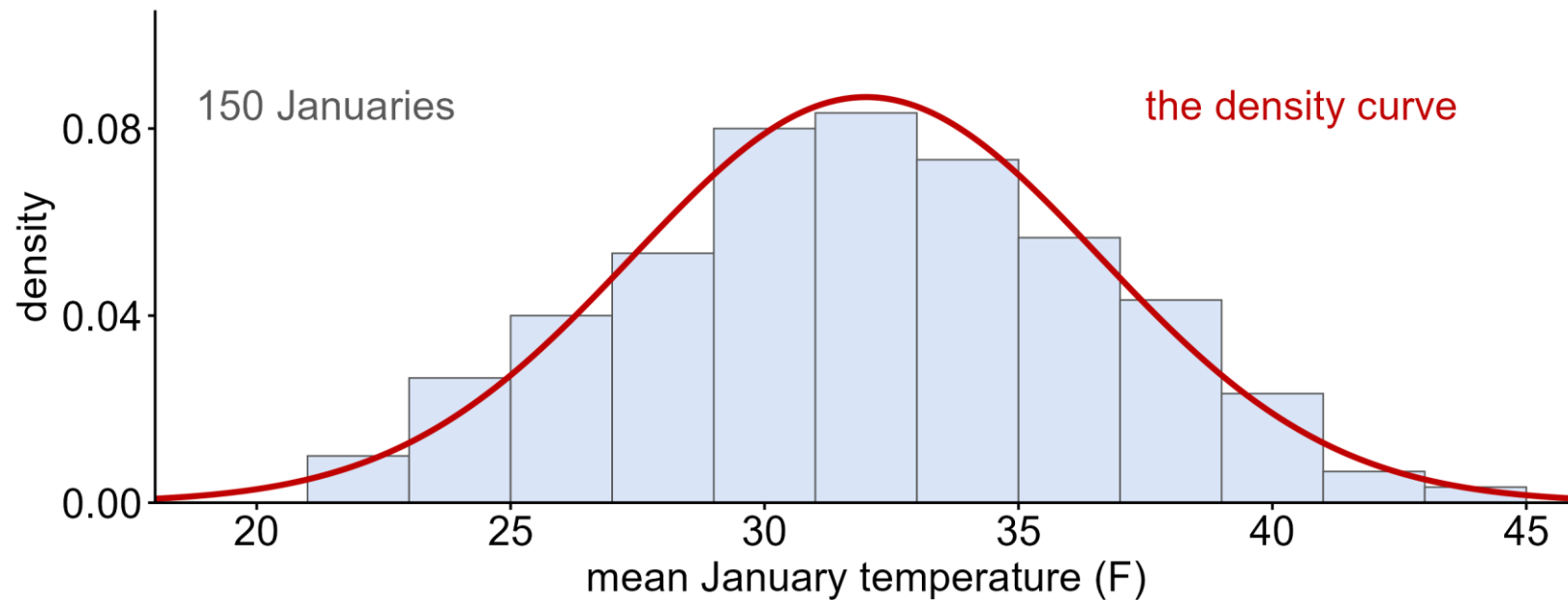
Averages of random values become normal, whatever the data look like

Most methods in this course are built on it

Density curves

density curve

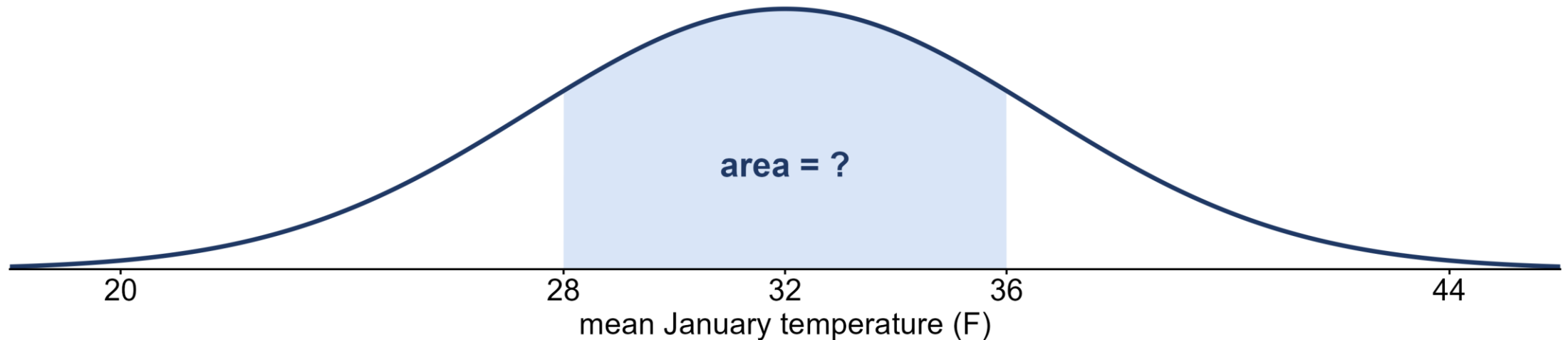
a smooth curve that describes a distribution; the total area under it is 1



Area under the curve is proportion

Central Park, 150 Januaries: $\bar{x} = 32.0^\circ\text{F}$ $s = 4.6^\circ\text{F}$

What fraction of Januaries averaged between 28°F and 36°F ?



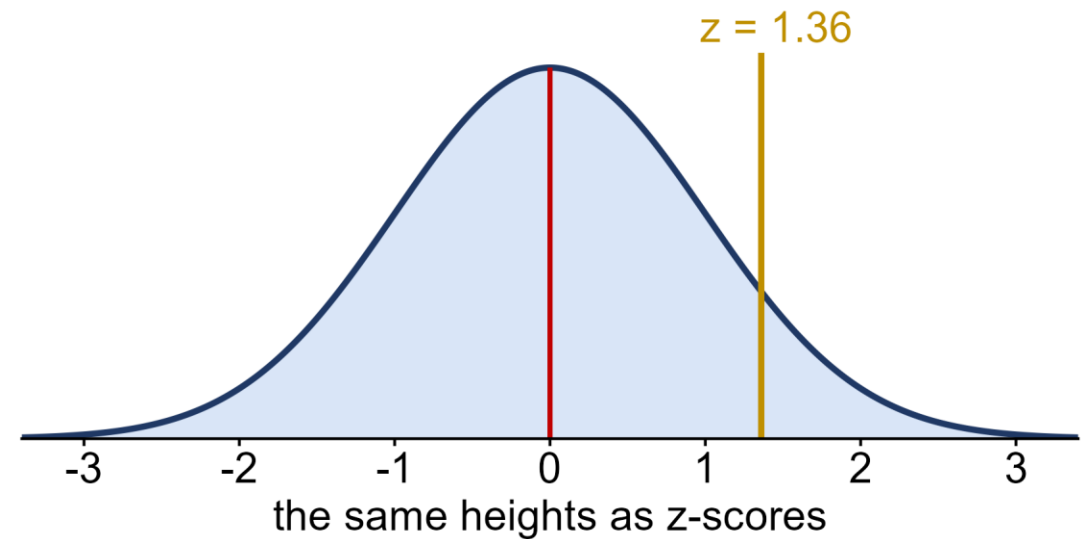
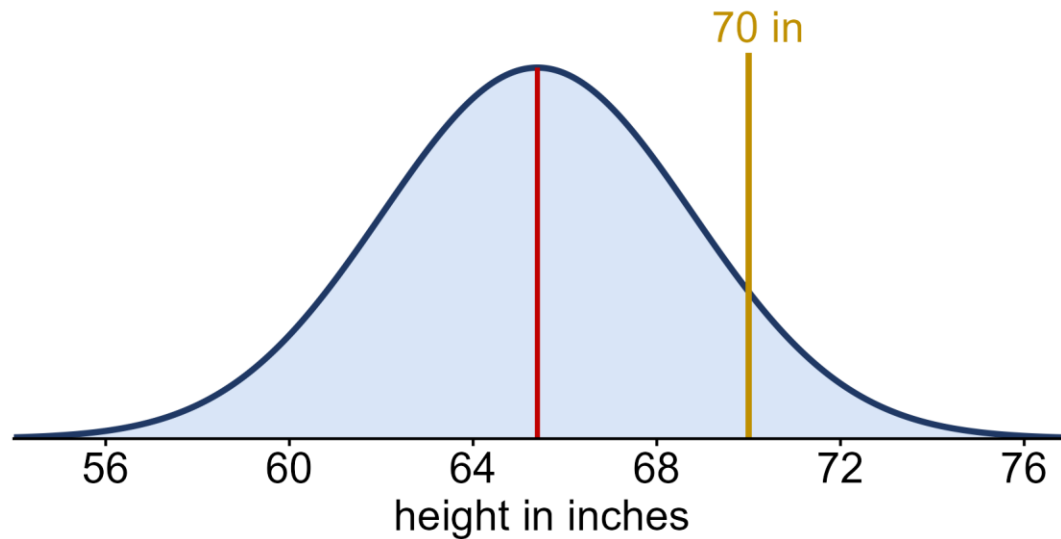
Area under the curve: 0.62 Actual: 92 of 150 = 0.61

The height of the curve is density: where values crowd. The area is the proportion.

The standard normal distribution

standard normal distribution

the normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$



Data: breakfast cereals

The twenty US breakfast cereals from Week 3, with their sodium and sugar content.

Y = the sodium content of a breakfast cereal, in mg

Cereal	Sodium	Sugar	Type
Frosted Mini Wheats	0	11	A
Raisin Bran	340	18	A
All Bran	70	5	A
Apple Jacks	140	14	C
Captain Crunch	200	12	C
...	...	...	...

n = 20 cereals

Which observation is more extreme?

X = an exam score (points) and Y = the sodium in a breakfast cereal (mg); each sample has $n = 20$.

Sample 1 (X): $\bar{x} = 74$, $s = 8$, observation $x = 90$

Sample 2 (Y): $\bar{y} = 167$, $s = 77$, observation $y = 290$

Which observation is more extreme?

$x = 90$: $16 / 8 = 2.0$ standard deviations above its mean

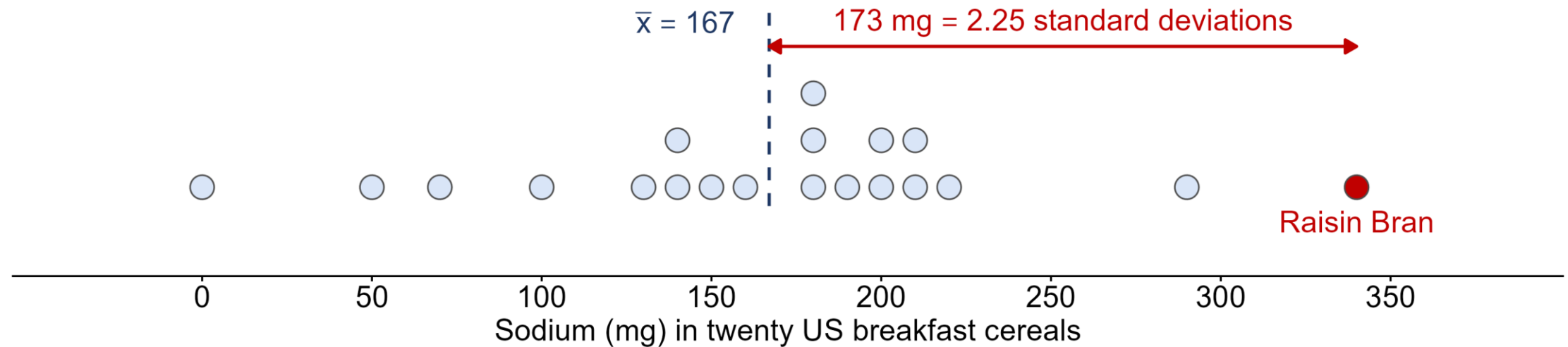
$y = 290$: $123 / 77 \approx 1.6$ standard deviations above its mean

$x = 90$ is more extreme

Measuring a deviation in standard deviations

How unusual is the sodium in Raisin Bran, among the 20 cereals from Week 3?

$x = 340 \text{ mg}$ $\bar{x} = 167 \text{ mg}$ $s = 77 \text{ mg}$



$$z = (x - \bar{x}) / s = 173 / 77 = 2.25 \rightarrow \text{the z-score}$$

Example: Jerome's points

Jerome averages 16 points a game, and one night he scores 10 (textbook Try It 6.2).

$$X \sim N(16, 4) \quad x = 10$$

$x = 10$ is ____ standard deviations to the ____ (right or left) of the mean ____.

$$z = (10 - 16) / 4 = -1.5$$

1.5 standard deviations to the left of the mean, 16

Data: EU carbon emissions

The World Bank's carbon dioxide emissions per person for the 28 member states of the European Union.

X = the carbon dioxide emissions per person in an EU country, in metric tons

Country Name	Country Code	CO2
Austria	AUT	8
Belgium	BEL	10
Bulgaria	BGR	6
Croatia	HRV	4.7
Cyprus	CYP	7
...	...	...

n = 28 countries

Example: carbon emissions across the EU

Carbon emissions per person in the 28 EU member states (World Bank).

$\bar{x} = 7.9 \text{ t}$ $s = 3.6 \text{ t}$ Luxembourg = 21.4 t

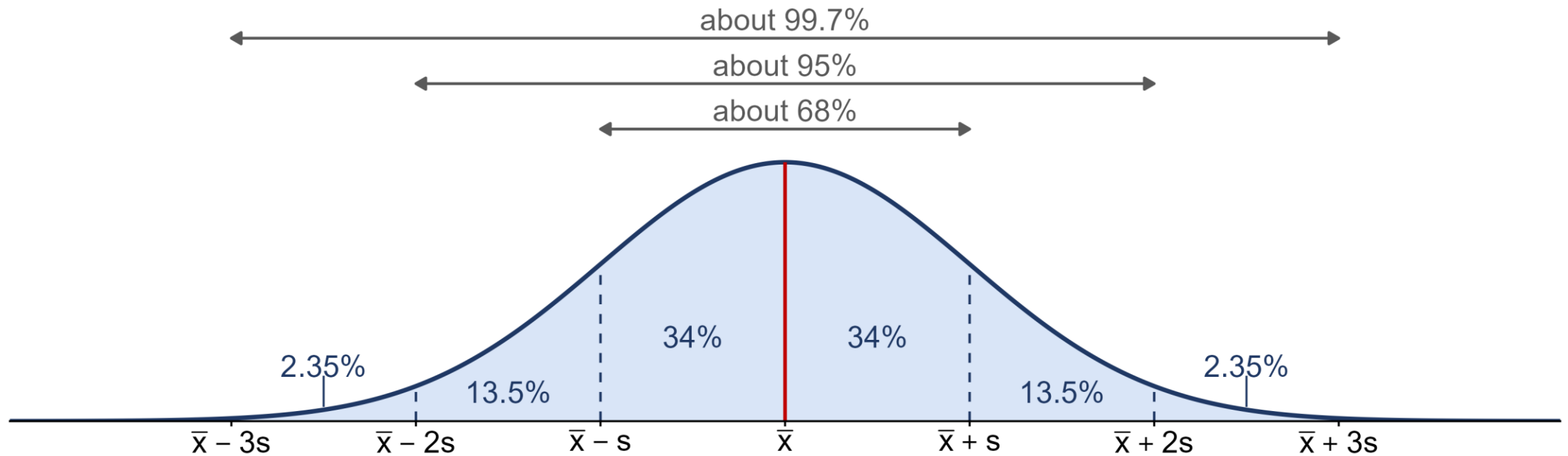
- (a) How many standard deviations above the mean is Luxembourg?
- (b) What emission level sits 2 standard deviations above the mean?

- (a) $z = 3.75$
- (b) 15.1 t, and only Luxembourg is above it

The empirical rule

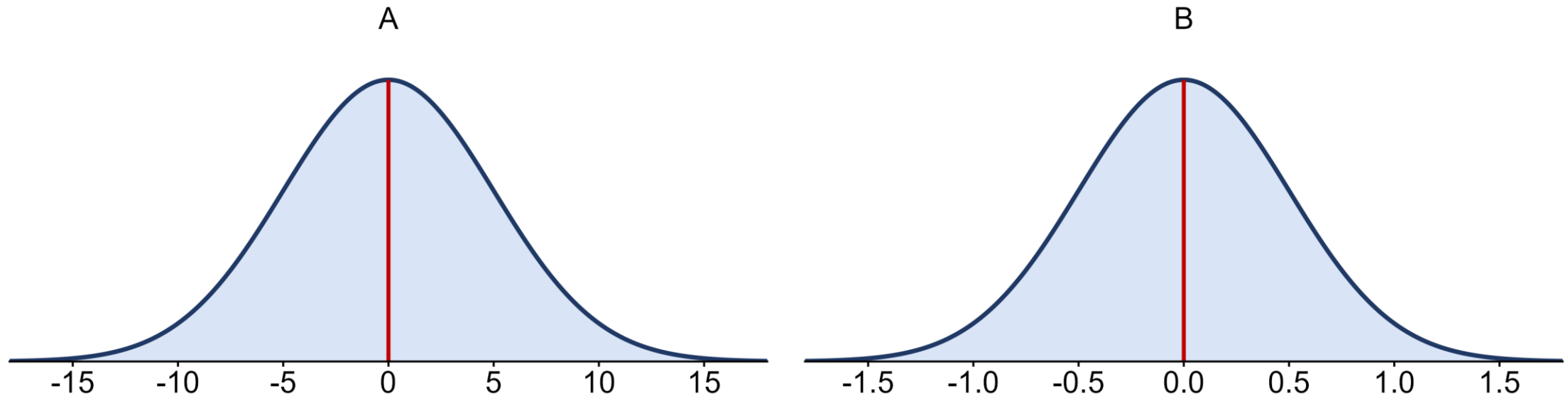
empirical rule

about 68% of values fall within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3



What is the standard deviation?

Estimate σ for curve A and for curve B.



A: $\sigma \approx 5$

B: $\sigma \approx 0.5$

Example: back to that exam

Exam scores in the class are roughly bell shaped.

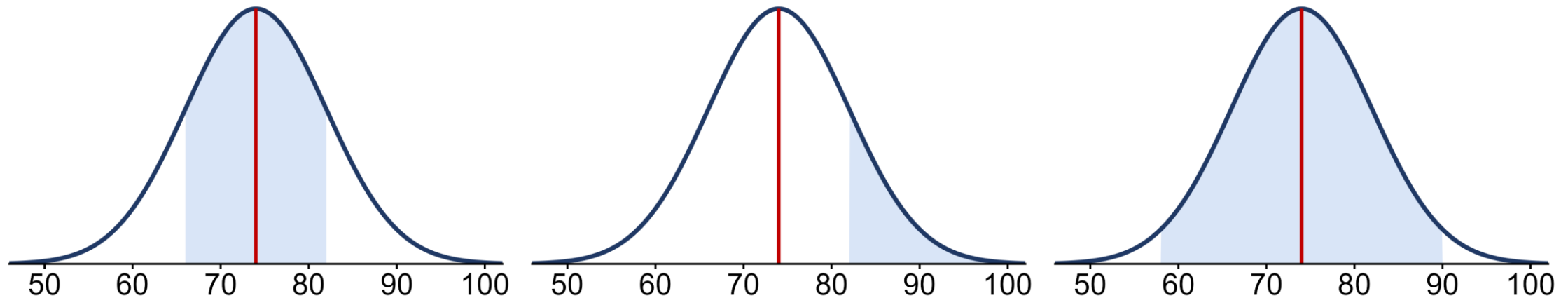
$\bar{x} = 74$ points $s = 8$ points

What percentage of scores falls in each shaded region?

(a) 66 to 82 points

(b) 82 points and above

(c) 58 to 90 points



(a) about 68%

(b) about 16%

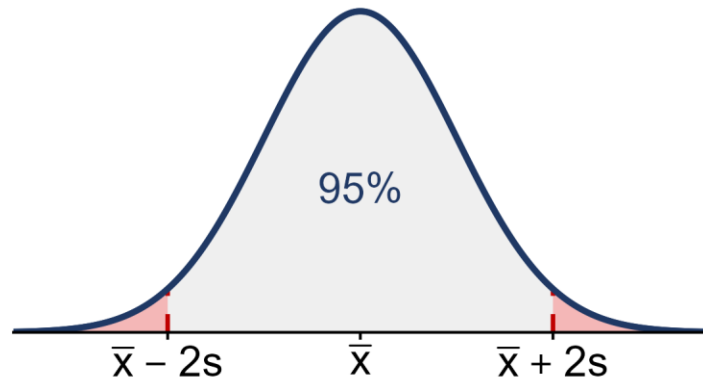
(c) about 95%

Outliers: the two-standard-deviation rule

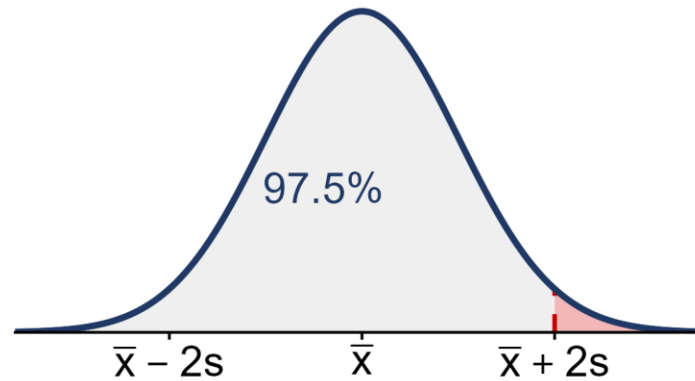
outlier

a value at least 2 standard deviations from the mean, $|z| \geq 2$ (about 5% of bell-shaped data)

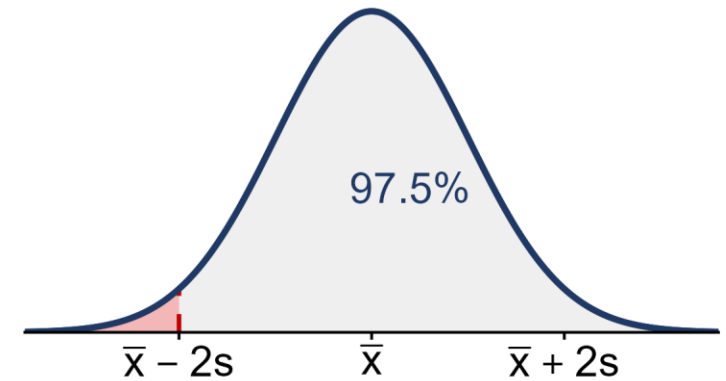
both tails: $2.5\% + 2.5\% = 5\%$



upper tail only: 2.5%

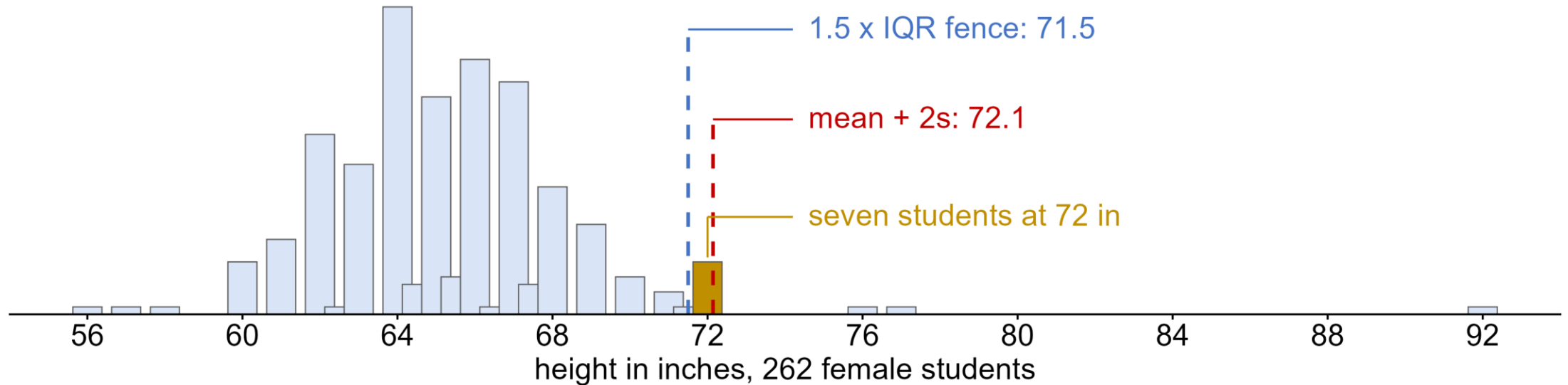


lower tail only: 2.5%



When the two outlier rules disagree

Do the $1.5 \times \text{IQR}$ rule and the $\pm 2s$ rule flag the same students?



No: the IQR rule flags 13 students, the $\pm 2s$ rule only 6

Why: the 92-inch value inflates s and pushes the $2s$ fence out; the quartiles barely move

Computing quantiles from a normal distribution

Heights of 262 female students are roughly bell shaped.

$\bar{x} = 65.4$ in $s = 3.38$ in one student is 70 in

- (a) What is her z-score?
- (b) What percentage of the women are shorter than she is?

- (a) $z = 1.36$
- (b) The empirical rule cannot say: 1.36 is not a whole number of standard deviations

How it used to be done: the Z-table

the standard normal table

each entry is the area to the LEFT of z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545

$z = 1.36$: row 1.3, column .06 → 0.9131, so about 91% are shorter

Today: the computer does the lookup

From a value to an area: pnorm

pnorm(1.36) = 0.9131

From an area to a value: qnorm

qnorm(0.90) = 1.28

No R in front of you?

The normal calculator at stat-tools.jmks-homelab.dev does both.

A note about transformations of variables

Changing units

We often need to change the units of a variable: °F to °C, feet to meters, dollars to euros.

Linear transformations: adding, subtracting, multiplying, dividing

$y = ax + b$ (a scales, b shifts). The z-score is a linear transformation. They preserve the shape of the distribution.

Nonlinear transformations: squaring, taking roots, logarithms, exponentiation

They do NOT preserve the shape of the distribution.

What a linear transformation does to the summaries

The center moves with the whole transformation

$\bar{y} = a\bar{x} + b$, and the same for the median.

The spread ignores the shift

s and IQR are multiplied by $|a|$; b has no effect.

The shape does not change at all

Skewness, gaps and outliers all survive unchanged.

Example: inches to centimeters

The 262 heights are converted from inches to centimeters: $y = 2.54x$.

$\bar{x} = 65.4$ in $s = 3.38$ in the 70-inch student = 177.8 cm

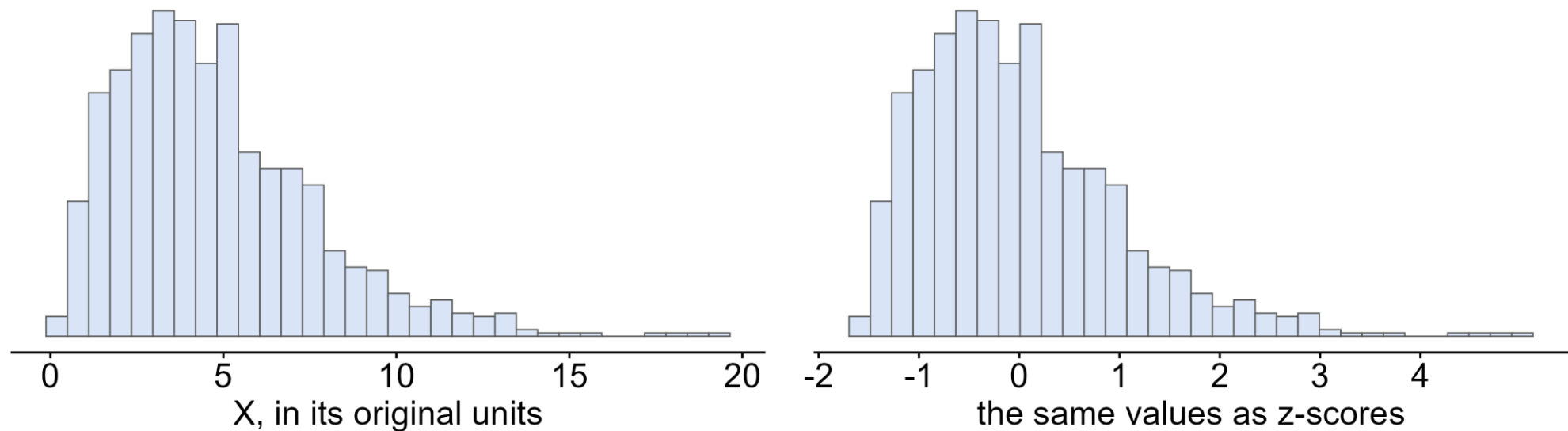
- (a) What are the mean and standard deviation in centimeters?
- (b) What is her z-score in centimeters?

(a) $\bar{y} = 166.1$ cm, $s = 8.59$ cm

(b) $z = 1.36$, unchanged

Does a z-score make data normal?

Does converting skewed data to z-scores make it normal?

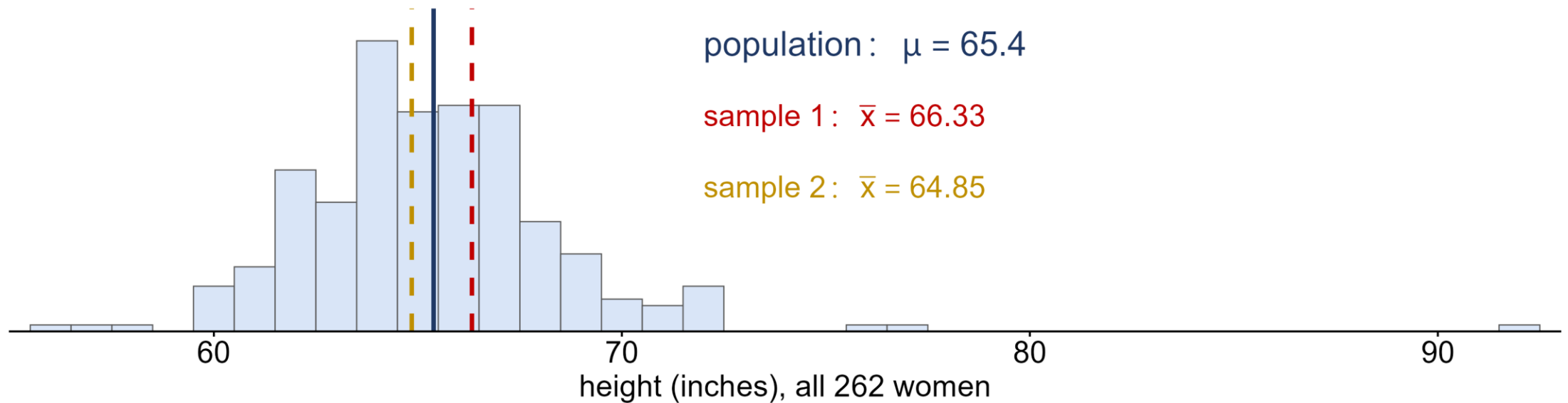


No: standardizing changes the axis, not the shape

Estimating a mean from a sample

Treat the 262 women as the population, and estimate μ from a random sample of $n = 20$.

Why do the two samples miss μ , and by different amounts?

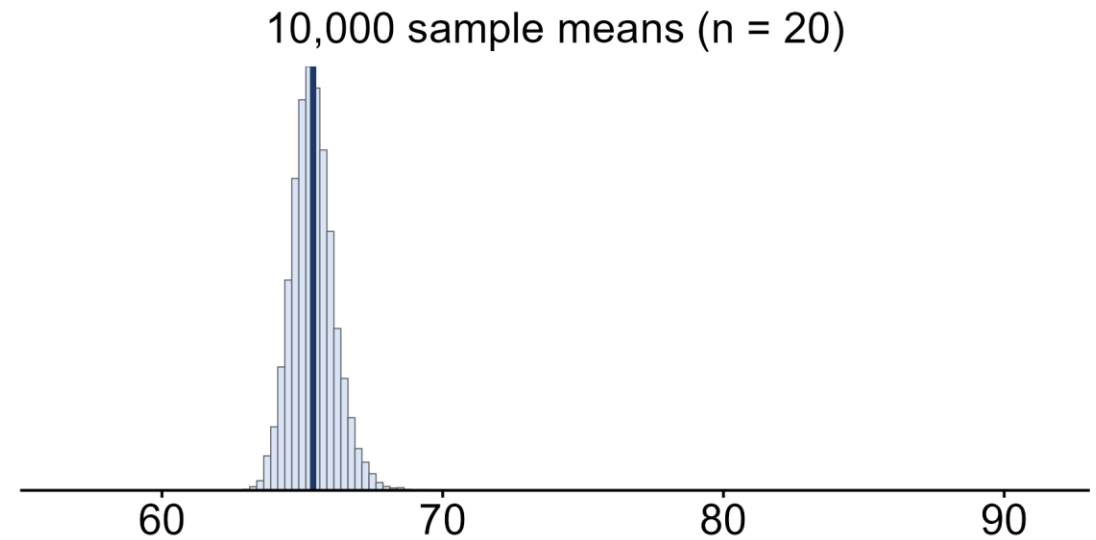
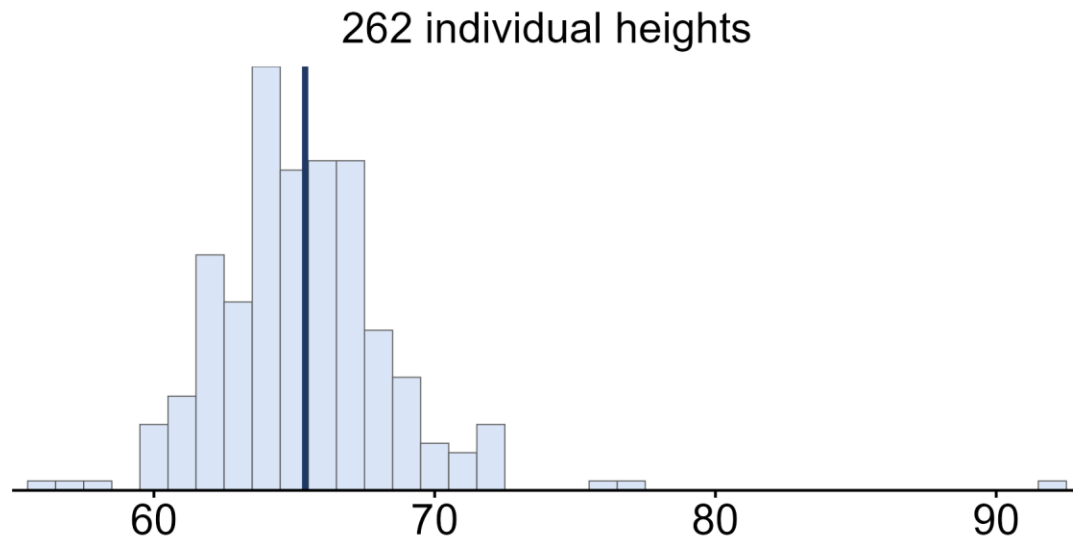


Each sample holds different students, so each $\bar{x}$ misses by a different amount: sampling error

Sampling distributions

sampling distribution

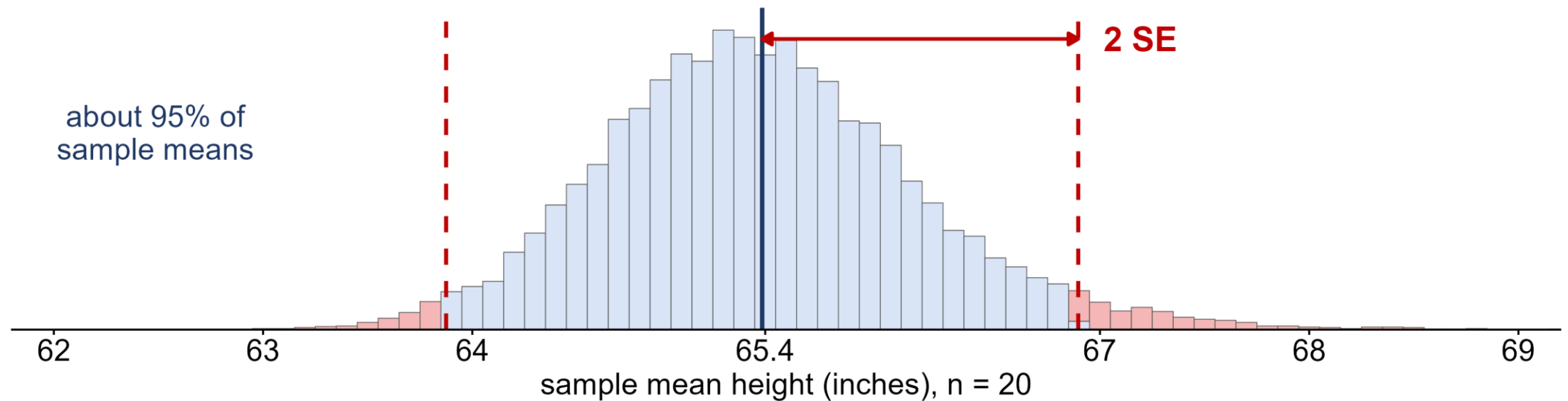
the distribution of a statistic, such as $\bar{x}$, over all possible samples of the same size n



Margin of error

margin of error

how far we expect an estimate to fall from the parameter: the largest distance to an estimate that is not an outlier, $2 \times \text{SE}$



Example: the margin of error for $\bar{x}$

Random samples of $n = 20$ from the 262 women.

$\sigma = 3.38$ in $n = 20$

What are the standard error and the margin of error of $\bar{x}$?

$$SE = \sigma / \sqrt{n} = 3.38 / \sqrt{20} \approx 0.76 \text{ in}$$

Margin of error = $2 \times 0.76 \approx 1.5$ in: about 95% of samples land within 65.4 ± 1.5 in

Statistical significance

Someone hands you a random sample of 20 students and says they are from our 262 women.

$\bar{x} = 70.33$ in $\mu = 65.4$ in $SE = 0.76$ in

Is this result statistically significant?

$z = (70.33 - 65.4) / 0.76 \approx 6.5$: far outside the margin of error, so yes
Statistically significant: too large to be ordinary sampling variation (they were men)

What you should know: the normal distribution

The two parameters

$x \sim N(\mu, \sigma)$: μ locates the center, σ sets the spread.

The empirical rule

68%, 95% and 99.7% within 1, 2 and 3 standard deviations (bell-shaped only).

The z-score

$z = (x - \bar{x}) / s$ counts standard deviations from the mean; $|z| \geq 2$ flags an outlier.